

Department of CS&IT
PROGRAMME: BCA (Bachelor of Computer Applications)-MACT
SEMESTER – V

Teaching Learning and Evaluation Plan

Course Information:

Course Code: 23BCA5M01

Course Title: Android Application Development

Credits Units: 03

Contact Hours: 45

L-T-P-E: 3-0-0-3

IA: UE Weightage - 50: 50

Pass Marks (IA & UE) – NA&18

Aggregate Pass Marks: 40

UE Question Paper Marks: 50

Special Examination Fees: NA

Pre-requisite (if any):

1. Students should know the programming language such as Java.
2. Students should know the basic concepts of Mobile Computing.

Course Facilitator (s): Dr. Taskeen Zaidi, Associate Professor, School of CS & IT

Programme Outcomes (POs) and Programme Specific Outcomes (PSOs)

At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well-informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentation and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economical, environmental

	social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11	Individual & Team Work: Ability to work as a member or leader in diverse teams in a multidisciplinary environment.
PO12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

PSO's

PSO1 Demonstrate a solid understanding of the basic concepts of Cloud Technologies and Mobile Computing.

PSO2 Recognize various cloud computing and mobile computing services.

PSO3 Use the latest tools and techniques of cloud and mobile computing.

PSO4 Design and develop solutions for real-world problems using Mobile and Cloud Computing Technologies

Course Objectives:

COB1.	To facilitate students to understand android SDK
COB2	To help students to gain a basic understanding of Android application development
COB3	To inculcate working knowledge of Android Studio development tool

Course Outcomes:

At the end of the course, students will be able to

Sl. No.	Course Outcome	Description	POs	Bloom's Taxonomy Level	Assessment Tool
1.	CO 1	Describe Advanced Android Development framework.	PO01, PO02, PO03 PO04 PO05	L2	MOOC COURSE/ Presentation/ IT1/Mini Project/ESE
2.	CO 2	Explain Mobile Data Management.	PO01, PO02, PO03 PO04 PO05	L4	IT1/ MOOC COURSE/ Preparatory /Mini Project/ ESE
3.	CO 3	Assess Native Capabilities for android application.	PO01, PO02, PO03 PO04 PO05	L5	IT2/ MOOC COURSE/ Mini Project/ ESE
4.	CO 4	Discuss architecture of flutter Application.	PO01, PO02, PO03 PO04 PO05	L2	IT2/ MOOC COURSE/ Mini Project/ ESE
5.	CO 5	Create android projects using android web services	PO01, PO02, PO03 PO04 PO05 ,PO8,PO11	L6	IT2/ MOOC COURSE /Report Writing on Mini project topic/application/ ESE

CO-PO/PSO Mapping:

Course Outcome	Bloom's Taxonomy	Program Outcomes (PO)	Program Specific Outcomes (PSO)
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	Level																
		PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PO 12	PS O 1	PS O 2	PSO 3 L4	PSO 4 L6
CO1	L2	3	3	3	3	3								2		1	2
CO2	L4	3	3	2	3	2								2	3	3	2
CO3	L5	3	3	3	3	3								2	3	3	2
CO4	L2	3	3	2	2	2								3	2	2	2
CO5	L6	3	2	3	3	2			3			3		3	3	3	3
CO Avg.		3	2.8	2.6	2.8	2.4			3			3		2.4	2.2	2.4	2.2

Course Contents:

Module	Details	Contact Hours
I	Module 1: Advanced Android Development Graphics and Animation, Work with 2D Graphics, Bitmap, Animation, Frame Animation, Tween Animation, View Animation, Multimedia in Android, Play Audio Files, Play Video Files Work in Background, Services, Notification Services, Broadcast Receiver	9
II	Module 2 Mobile Data Management Content Providers: Contents provider, Uri, CRUD access, Browser, CallLog, Contacts, Media Store, and Setting. Data Access and Storage: Shared Preferences, Storage External, Network Connection. SQLite - SQLite Databases, Introduction to Firebase with simple CRUD Operation	9
III	Module 3: Native Capabilities Camera, Audio, Sensors and Bluetooth: Android Media API: API uses, Playing audio/video, Media recording. Sensors -how sensors work, listening to sensor readings. Maps & Location: Android Communications: GPS, Working with Location Manager, Working with Google Maps extensions, Maps via intent and Map Activity, Location based Services. Location Updates, location-based services (LBS), Location Providers, Selecting a Location Provider, Finding Your Location, Map - Based Activities, How to load maps, To finding map API key.	9
IV	Module4: Flutter Introduction to Flutter, Features of Flutter, Advantages of Flutter, Disadvantages of Flutter, Flutter installation, Architecture of flutter application, creating simple application in android studio, Android Firebase Authentication.	9

V	Module 5:Android Web Services Uses and characteristics of Json in Real time, Json Parser, UI Thread, Async Task, Operations (onpreExecute(), doInBackground(),onPostExecute()), API formation, Android Application to parse Json Data from API, Design and Develop Web services and Parsing, JSON Parsing, Access web data with JSON, Connect to Web Services, Using Async Task & Third Party Library : Retrofit and volle	9
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TEXT BOOKS: (Write in APA Format)

Text Books:	
TB1	John Horton,(2021),“ Android Programming for Beginners”,3rd Edition, Packt Publishing
TB2	ThomasBailey , Alessandro Biesse,(2021) “Flutter for Beginners: An Introductory Guide to Building Cross-platform Mobile Applications with Flutter 2.5 and Dart”, 2nd Edition, Packt Publishing, 2021
References Books:	
1	Erik Hellman, “Android Programming – Pushing the Limits”, 1st Edition, Wiley India Pvt Ltd
2	J F DiMarzio, “Beginning Android Programming with Android Studio”, 4th Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978- 8126565580
3	https://www.tutorialspoint.com/flutter/flutter_tutorial.pdf

Session-Wise Plan:

Pedagogy/Activity planned		Mode of Delivery	
P1	PPT presentation & Classroom interaction	M1	Synchronous – PPT
P2	Blended Learning with Hands-on	M2	Asynchronous/synchronous
A1	Activity-1:MOOC	M3	Synchronous - Hands-on
A2	Activity-2 Mini Project	M4	Synchronous - Discussion

Relevant Website (RW):

RW-1	Animation in Android with Example: https://www.geeksforgeeks.org/animation-in-android-with-example
RW-2	How To Perform CRUD Operations SQLite In Android: https://vasundhara.io/blogs/crud-operations-sqlite-in-Android
RW-3	How to get user location in Android: https://www.geeksforgeeks.org/how-to-get-user-location-in-android/
RW-4	Add Firebase to your Android project https://firebase.google.com/docs/android/setup

RW-5	Andriod Application Design with theme and styles - https:// www.vogella.com/tutorial Using Async Task & Third Party Library : Retrofit and volleys/AndroidStylesThemes/article.html
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Video Links (VL):

VL-1	Introduction to Android Animation - https://www.youtube.com/watch?v=Pz95C_N0ike
VL-2	How to play sound file in your app https://www.youtube.com/watch?v=0SJAK_pKUoE
VL-3	Content Provider to Get Contacts Android Example - https://www.youtube.com/watch?v=UuUikmsBr_M
VL-4	Android Get Current Location - https://www.youtube.com/watch?v=qY-xFxZ7HKY
VL-5	Saving Data - I - https:// www.youtube.com/watch?v=oArUFc2mOHY&list=PL49hKkxjHVqHilv8CUB-p9gMPFZtfqZTN&index=28
VL-6	Security and Permissions - https://www.youtube.com/watch?v=AvRhT6Jx-2Y&list=PL49hKkxjHVqHilv8CUB-p9gMPFZtfqZTN&index=30
VL-7	Services - https:// www.youtube.com/watch?v=mhJKKAWZvgQ&list=PL49hKkxjHVqHilv8CUB-p9gMPFZtfqZTN&index=31

MOOC Courses (MC):

Sr.No.	Platform	Topic	CO	Link
MC-1	Coursera	Build Your First Android App (Project-Centered Course)	CO1 CO3 CO5	https://www.coursera.org/learn/android-app
MC-2	Coursera	Meta Android UI Development Specialization	CO2 CO3 CO5	https://www.coursera.org/specializations/meta-android-ui-development
MC-4	LinkedIn Learning	Flutter Essential Training: Build for Multiple Platforms	CO4	https://www.linkedin.com/learning/flutter-essential-training-build-for-multiple-platforms/building-for-multiple-platforms?u=92695330
Total MOOC Course integration with Certification				

Sr.No.	Platform	Topic	CO	Link	Duration
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MC-1	Course era	Build Your First Android App	C01 -C05	https://www.coursera.org/learn/android-app	21Hrs
MC-2	Course era	Flutter and Dart: Developing iOS, Android, and Mobile Apps	C04	https://www.coursera.org/learn/flutter-and-dart-developing-ios-android-mobile-apps	20Hrs

Hardware/Software/Equipment (HSE):

HSE-1	Andriod studio by Google - https:// developer.android.com/studio
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Session-wise Planning:

Module	Session	Topic	TB/RB/ OR/R W/VL/	Pedagogy/ Activity planned	CO	Mode of delivery
Bride Course	Zero Lecture	AAD- Need, Scope, Opportunity & Career+TLEP Discussion	VL-1 to VL-5	Blended Learning Approach		Synchronous-PPT Asynchronous- Video
I	1	Graphics and Animation	TB-1 VL1 RW1 RB1	PPT presentation & Classroom interaction	CO 1	Synchronous-PPT Asynchronous- Video
	2	Work with 2D Graphics, Bitmap, Animation	TB-1 RW1 VL-1 RB-1	PPT presentation & Classroom interaction		Interactive Lecture Synchronous-Video
	3	Frame Animation, Tween Animation,	TB-1 VL1 RW1 RB-1	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	4	View Animation, Multimedia in Android	TB-1 VL-1 RW1 RB-1	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video

	5	Play Audio Files	TB-1 VL-2 RB-1	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	6	Play Video Files, Work in Background	TB-1 RB-1	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	7-8	Services, Notification Services, Broadcast Receivers	TB-1 RB-1	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	9	Activity-1- Creating a Mobile App Interface in After Effects	MC-1	Interactive & Assessment with Report		Synchronous-Discussion, Suggestions Assessment Certification on Completion
II	10	Contents provider, Uri, CRUD access	TB-1 VL-3 RW2 RB-2	PPT presentation & Classroom interaction	CO 2	Synchronous-Interactive Video Lecture with Quiz
	11	Browser,	TB-1 RB-2	PPT presentation & Classroom interaction		Synchronous-Interactive Video Lecture
	12	Call Log, Contacts	TB-1 VL-3 RB-2	PPT presentation & Classroom interaction		Synchronous - PPT
	13	Media Store and Setting	TB-1 RB-2	PPT presentation & Classroom interaction		Synchronous - PPT
	14	Data Access and Storage: Shared Preferences,	TB-1 VL-5 RW2 RB-2	PPT presentation & Classroom interaction		Synchronous - PPT
	15	Storage External, Network Connection	TB-1 RB-1 VL3	PPT presentation & Classroom interaction		Synchronous-Interactive Video Lecture

	16	SQLite - SQLite Databases	TB-1 RW2 RB-2	Open & Close Discussion, Query Resolution, Monitoring		Synchronous-PPT Asynchronous- Video
	17	Introduction to Firebase with simple CRUD Operation	TB-1 RW-4 RB-2	PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	18	ACTIVITY1: Android Development Essential Training: 4 Working with Data	MC-2	PPT Presentation & Classroom interaction		Synchronous-Discussion, Suggestions Assessment Certification on Completion
III	19	Camera, Audio, Sensors and Bluetooth: Android Media API: Playing audio/video	RW-3 TB-1 RB-2	PPT Presentation & Demonstration	CO 3	Synchronous-PPT Asynchronous- Video
	20	Media recording. Sensors - how sensors work, listening to sensor readings	TB-1 RW-3 RB-2	PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	21	Bluetooth. Maps & Location: Android Communications: GPS	TB-1 RW-3 VL-4 RB-2	PPT Presentation & Demonstration		Synchronous-PPT Asynchronous- Video
	22	Working with Location Manager, Working with Google Maps extensions	TB-1 RW-3 VL-4 RB-2	PPT Presentation & Classroom interaction		Synchronous-PPT Lecture
	23	Maps via intent and Map Activity, Location based Services. Location Updates, location-based services (LBS)	TB-1 RW-3 VL-4 RB-2	PPT Presentation & Demonstration		Synchronous-PPT Asynchronous- Video
	24	Location Providers, Selecting a Location Provider, Finding Your Location	TB-1 RW-3 VL-4 RB-2	PPT Presentation with Comparative Analysis of IDS & NIDS		Synchronous-PPT Asynchronous- Video
	25	Map - Based Activities,	TB-1 RW-3 VL-4	PPT Presentation & Demonstration		Synchronous-PPT Asynchronous- Quiz

			RB-2			
	26	How to load maps, To finding map API key	TB-1 RW-3 RB-2	PPT presentation & Classroom interaction		Synchronous-PPT & Interactive Video
	27	ACTIVITY1: Integrating GPS and Points of Interest	MC-3	Interactive & Assessment with Report		Assessment Certification with Report
IV	28	Introduction to Flutter,	TB-2 RB-3	PPT Presentation & Classroom interaction	CO 4	Synchronous-PPT Lecture with interaction
	29	Features of Flutter, Advantages of Flutter,	TB-2 RB-3	PPT Presentation & Demonstration		Synchronous-PPT Lecture with demo
	30	Disadvantages of Flutter, Flutter installation,	TB-2 RB-3	PPT Presentation & Classroom Interaction		Synchronous-PPT Asynchronous- Video
	31	Architecture of flutter application	TB-2 RB-3	PPT Presentation & Comparative Analysis		Synchronous-PPT Asynchronous- Video
	32	creating simple application in android studio	TB-2 RB-3	Classroom interaction with live demo		Synchronous-PPT & Interactive Video with Quiz
	33	Android Firebase Authentication	TB-2 RB-3	PPT Presentation & Classroom Interaction		Synchronous-PPT Asynchronous- Video
	34	Android Firebase Authentication	TB2 RW-4 RB-3	Classroom discussion		Synchronous-Discussion, Suggestions Clarification
	35	Activity-1- Flutter Essential Training: Build for Multiple Platforms	MC-4	Interactive & Assessment with Report		Assessment Certification with Report
	36	ACTIVITY : Code Implementation and Testing		Open & Close Discussion, Query		PPT Assessment/Execution/ Report

				Resolution, Monitoring		Submission
V	37	Uses and characteristics of Json in Real time, Json Parser	TB-1 RW-5 VL-6 RB-1	PPT & Explanation	CO 5	Synchronous- PPT Presentation
	38	UI Thread, Async Task, Operations (onpreExecute()),	TB-2 RW-5 VL-6 RB-1	PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	39	dolnBackground(),onPostExecute(),	TB-1 RW-5 VL-6 RB-1	Video Demonstration with Student Interaction		Synchronous-PPT & Interactive Video with Quiz
	40	API formation, Android Application to parse Json Data from API	TB-2 RW-5 VL-7 RB-1	Video Demonstration with Student Interaction		Synchronous-PPT & Interactive Video
	41	Design and Develop Web services and Parsing, JSON Parsing,	VL-7 RW-5 RB-1	PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	42	Access web data with JSON	VL-7 RW-5 RB-1	PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	43	Connect to Web Services , Using Async Task & Third Party Library : Retrofit and volley	TB-2 RW-5 RB-1	PPT Presentation & Classroom interaction		Synchronous live interaction with student and sharing and suggesting writing skills for reports.
	44	Activity1: Android Development: Retrofit with Java	MC-5	Interactive & Assessment with Report		Assessment Certification with Report
	45	Activity2Mini Project App review on Andriod Studio		Open & Close Discussion, Query Resolution, Monitoring		Asynchronous- Video, PPT Assessment Certification on Completion

Assessment Scheme: CA:ESE - 50:50

Assessment Scheme: IA: UE - 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency	Weightage (%)	CO
1.	Activity 1-MOOC	Formative	1	15	CO1,CO2,CO3,CO4,CO5
2.	Activity 2-Mini Project		1	15	CO1,CO2,CO3,CO4,CO5
3.	Internal Test1		1	5	CO1, CO2,CO3
4.	Internal Test2	Formative	1	10	CO3,CO4,CO5
5.	Class participation		continuous	5	**5 MARKS based on class participation, activities submission on time and test attendance
6.	End Semester Exam	Summative	1	50	CO1, CO2, CO3, CO4,CO5
	Total			100	

Others Details:

- Class Participation – Punctuality, Frequency, and Level of Interaction, Contribution to the Learning Process (1 – Poor/Below 75%, 2 – Good/75-80%, 3 – Very Good/80-90%, 4 and above – Excellent/Above 90%)
- Review Paper – Parameters for Evaluation:
 - Stated objectives
 - Provides evidence
 - Critical analysis of evidence
 - Global self awareness
 - Perspective taking
 - Cultural diversity
- Discipline – The students are expected to be punctual and regular in attending classes, submitting assignments and activities. They should be decently dressed all the time. In their interactions, they should be respectful and sensitive to everyone.
- Chamber Consultation Hours – 11.00 am to 12.00 noon (Tentative)
- Pre Reading – The readings/ references given in session-wise plan should be read by students before coming to class to have effective and meaningful discussion on the topic.
- Participation in Co-curricular and Extra-Curricular Activities – Attendance for the course on Multidisciplinary Writing is compulsory as it will help in writing Review Essays

Bloom's Taxonomy-Revised

LEVEL	DESCRIPTION	MEANING	ACTION VERBS
6	Creating	Can the student create a new product or POV?	Assemble, construct, create, change, combine, compose, design, develop, formulate, invent, modify, organize, propose, theorize, write
5	Evaluating	Can the student justify a stand or decision?	Appraise, agree, assess, argue, conclude, decide, defend, judge, prioritize, prove, rate, recommend, select, support, value
4	Analyzing	Can the student distinguish between different parts?	Contrast, compare, criticize, differentiate, discriminate, dissect, distinguish, examine, experiment, operate, question, simplify, test
3	Applying	Can the student use information in a new way?	Choose, demonstrate, dramatize, employ, illustrate, interpret, schedule, sketch, solve, use
2	Understanding	Can the student explain ideas and concepts?	Classify, describe, discuss, explain, identify, infer, locate, outline paraphrase, recognize, report, summarize, select, translate
1	Remembering	Can the student recall or remember information?	Define, duplicate, find, list, label, match, memorize, name, omit, recall, repeat, state, spell, tell

Signature of the Faculty

Signature of the HoD